HW 3 posted tomorrow
due Wed 10/15

$$\begin{cases} u_t - \Delta u = f(x,t) & t>0,\ x\in\mathbb{R}^n \\ u(x,0) = g(x) \end{cases}$$

$$\Phi(x,t) = \frac{e^{-|x|^2/4t}}{(4\pi t)^{n/2}}$$

If $f=0$, solution formula is $u(x,t) = \int_{\mathbb{R}^n} \Phi(x-y,t)\, g(y)\, dy$

If $g=0$, solution formula is

$$u(x,t) = \int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s)\, f(y,s)\, dy\, ds$$

In general, formula is

$$\boxed{u(x,t) = \underbrace{\int_{\mathbb{R}^n} \Phi(x-y,t)\, g(y)\, dy}_{u_1} + \underbrace{\int_0^t \int_{\mathbb{R}^n} \Phi(x-y,t-s)\, f(y,s)\, dy\, ds}_{u_2} = u_1 + u_2}$$

$$\partial_t u_1 - \Delta u_1 = 0$$

$$\partial_t u_2 - \Delta u_2 = f$$

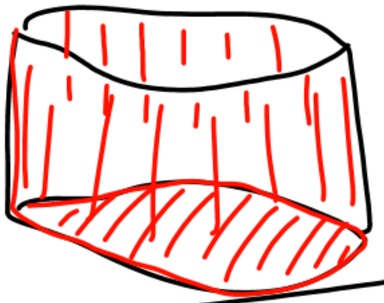
$$(\partial_t - \Delta)(u_1 + u_2) = (\partial_t - \Delta)u_1 + (\partial_t - \Delta)u_2$$
$$= 0 + f = f.$$
$$u_1(x,0) + u_2(x,0) = g + 0 = g$$

Let $U \subseteq \mathbb{R}^n$ be a bounded domain.
Consider heat flow on $U_T = U \times [0,T]$, a parabolic cylinder.

Q: How to get a well-defined problem?
i.e. where to specify boundary conditions?

A: Specify u on the <u>parabolic boundary</u> of U_T,

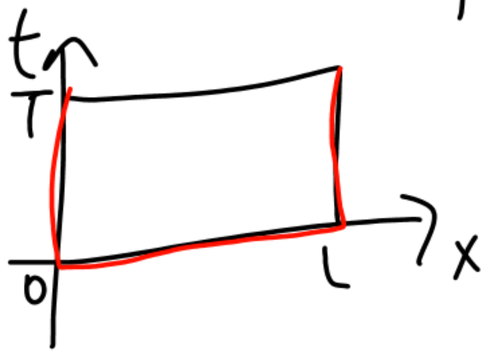
$$\Gamma U_T = \{(x,0) : x \in U\} \cup \{(x,t) : x \in \partial U,\ 0 \le t < T\}$$



Does not include $t = T$ slice!

1D case: $U = (0, L)$, $U_T = (0,L) \times [0,T]$

$$\Gamma U_T = ([0,L] \times \{0\}) \cup (0 \times (0,T)) \cup (L \times (0,T))$$

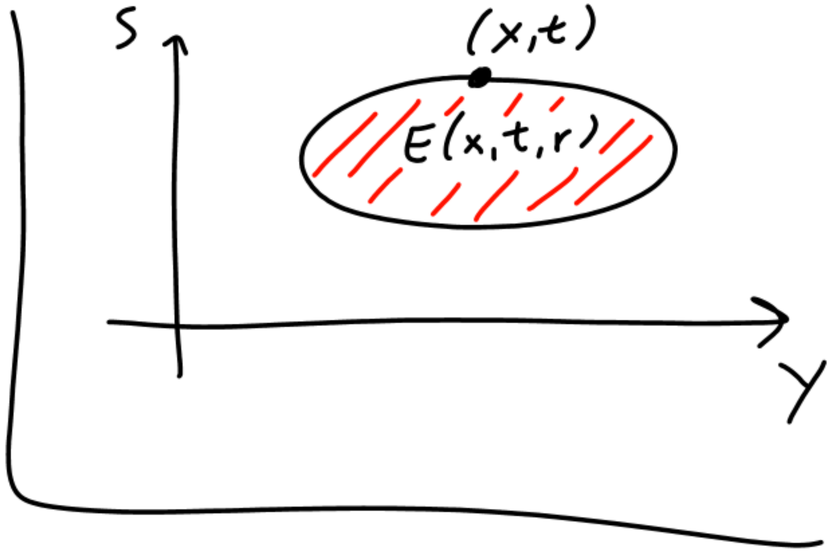


Specify BC on both sides, unlike transport equation.

(Recall: If $\Delta u = 0$ in $U \subseteq \mathbb{R}^n$ and $\overline{B(x,r)} \subseteq U$, then

$\Phi(x) = |x-y|^{2-n} c_n$

$$u(x) = \fint_{B(x,r)} u \, dy .)$$



To state mean value property for Heat Equation, define

$$E(x,t,r) = \left\{ (y,s) \in \mathbb{R}^n \times \mathbb{R} : s \le t \text{ and } \Phi(x-y, t-s) \geq \frac{1}{r^n} \right\}$$

<u>Thm</u> If $u_t - \Delta u = 0$ in U_T and $\overline{E(x,t,r)} \subseteq U_T$.

then $$u(x,t) = \iint_{E(x,t,r)} u(y,s) \frac{|x-y|^2}{4r^n (t-s)^2} \, dy \, ds$$

Maximum Principle

Look at heat equation on U_T:

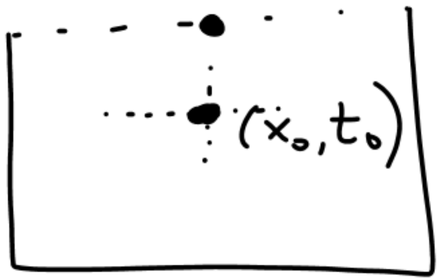
Thm Let $u \in C^1_t C^2_x(U_T) \cap C(\bar{U}_T)$ be a solution to $u_t - \Delta u = 0$ in U_T.

Then $\max_{\bar{U}_T} u = \max_{\Gamma U_T} u$.

Pf Maximum over $\bar{U}_T$ exists by compactness.

Assume max is achieved at an interior point $(x_0, t_0) \in \bar{U}_T \setminus \Gamma U_T$

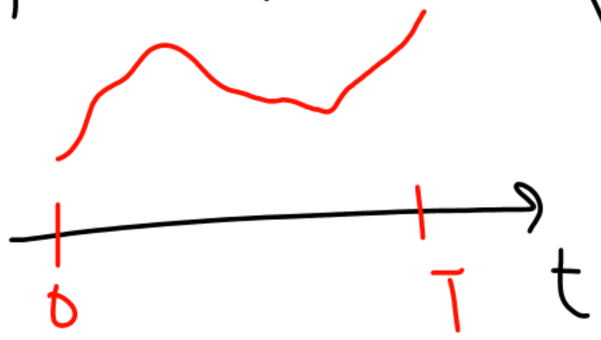
Idea: At a max point, $D_x^2 u(x_0, t_0) \leq 0$



$$\Delta u = \operatorname{tr}(D_x^2 u(x_0, t_0)) \leq 0$$

$u_t(x_0, t_0) = 0$? No, we only have $u_t(x_0, t_0) \geq 0$ because t_0 could equal T.

(Fix x,



)

$\Rightarrow u_t - \Delta u \geq 0$ at (x_0, t_0)

To make this rigorous, let $u_\varepsilon = u + \varepsilon |x|^2$

Then $(u_\varepsilon)_t - \Delta u_\varepsilon = u_t - \Delta u - 2n\varepsilon = -2n\varepsilon$

By above argument applied to u_ε, $-2n\varepsilon = (u_\varepsilon)_t - \Delta u_\varepsilon \geq 0$, a contradiction.

u_ε must achieve max on ΓU_T

$$\Rightarrow \max_{\overline{U}_T} u_\varepsilon = \max_{\Gamma U_T} u_\varepsilon .$$

$$u \le \underbrace{u + \varepsilon |x|^2}_{= u_\varepsilon}$$

$$\Rightarrow \max_{\overline{U}_T} u \le \max_{\overline{U}_T} u_\varepsilon = \max_{\Gamma U_T} u_\varepsilon$$

$$= \max_{\Gamma U_T} (u + \varepsilon |x|^2)$$

$$\le \max_{\Gamma U_T} u + \varepsilon M^2$$

M depends on U

Send $\varepsilon \to 0$. ✓

<u>Thm</u> There exists at most one solution in

$$C_t^1 C_x^2(U_T) \cap C(\bar{U}_T)$$

of

$$\begin{cases} u_t - \Delta u = f & \text{in } U_T \\ u(x,0) = g(x,0) & \text{in } U \times \{t=0\} \\ u(x,t) = h(x,t) & \text{in } \partial U \times [0,T] \end{cases}$$

<u>Pf</u> u & v both solutions, let $w = u - v$

$\Rightarrow w_t - \Delta w = 0$, $w = 0$ on ΓU_T. Maximum principle

to w & $-w$ $\Rightarrow w = 0$, so $u = v$.

Maximum Principle on $\mathbb{R}^n$:

<u>Theorem</u> If u is a <u>bounded</u> solution to $u_t - \Delta u = 0$ in $\mathbb{R}^n \times [0,T]$ and $u \in C^1_t C^2_x(\mathbb{R}^n \times [0,T])$, then

$$\max_{\mathbb{R}^n \times [0,T]} u = \max_{\mathbb{R}^n} u(x,0)$$

(By more careful arguments, one can recover this conclusion if $|u(x,t)| \leq A e^{a|x|^2}$ for some $a, A > 0$.)

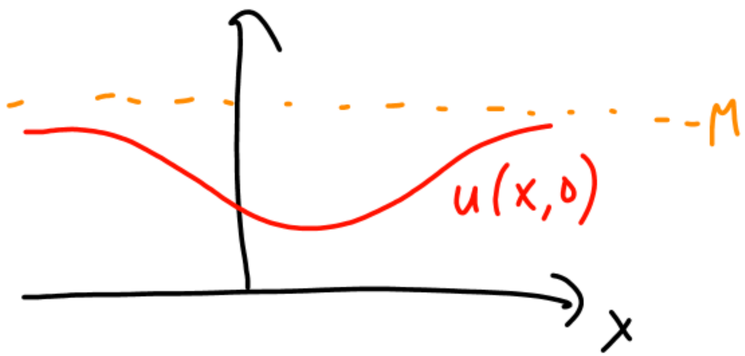
Proof for bounded u:

Before, on a bounded domain, we could find a maximum point by compactness.

This doesn't hold on $\mathbb{R}^n \times [0,T]$.

Define $v(x,t) = \varepsilon\left(t + \frac{|x|^2}{2n}\right) + M$,

where $u(x,0) \le M$ for all $x \in \mathbb{R}^n$.



Claim: $v_t - \Delta v = 0$

$$v_t = \varepsilon, \quad \Delta v = \frac{\varepsilon}{2n}\Delta|x|^2 = \frac{\varepsilon}{2n}\sum_{i=1}^{n}\partial_{x_i}^2 |x|^2 = \varepsilon$$

✓

Claim: for $R > 0$ sufficiently large,

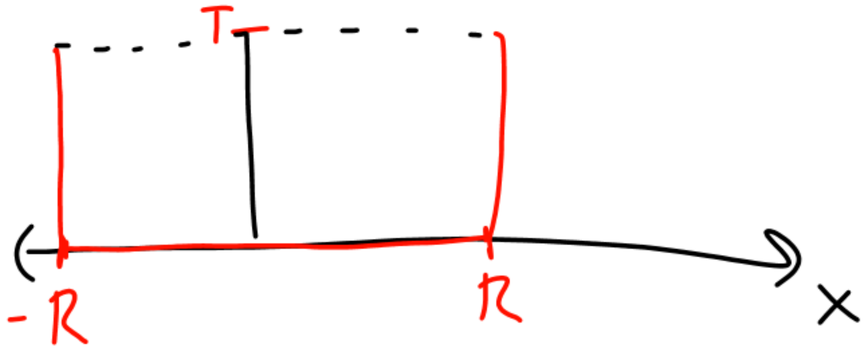
$$u \le v \quad \text{on} \quad \Gamma\left(B(0,R) \times [0,T]\right)$$

When $t = 0$ and $|x| \le R$,

$$v(x,t) = \varepsilon\left(t + \frac{|x|^2}{2n}\right) + M$$

$$v(x,0) = \frac{\varepsilon |x|^2}{2n} + M$$

$$\le \frac{\varepsilon R^2}{2n} + M$$



$$u(x,0) \le M \le \varepsilon\left(\frac{|x|^2}{2n}\right) + M$$

$$= v(x,0) \quad \checkmark$$

If $|x| = R$,

$$v(x,t) = \varepsilon\left(t + \frac{R^2}{2n}\right) + M \ge A \ge u$$

(R large enough)

Since $u-v$ solves heat equation

and $u-v \le 0$ on $\Gamma(B(0,R)\times[0,T])$,

max principle on $B(0,R)\times[0,T]$

implies $u-v \le 0$ in $B(0,R)\times[0,T]$

$$\Rightarrow \quad u(x,t) \le \varepsilon\left(t+\frac{|x|^2}{2n}\right)+M, \quad |x|\le R$$

Send $R\to\infty$: $u(x,t) \le \varepsilon\left(t+\frac{|x|^2}{2n}\right)+M, \quad x\in\mathbb{R}^n$

Send $\varepsilon\to 0$: $u(x,t)\le M$

M is any upper bound for $u(x,0) \Rightarrow u(x,t) \le \sup_{\mathbb{R}^n} u(x,0)$.

<u>Thm</u> There is at most one bounded solution to

$$\begin{cases} u_t - \Delta u = f & \text{on } [0,T] \times \mathbb{R}^n \\ u(x,0) = g(x) \end{cases}$$

if g is bounded.